

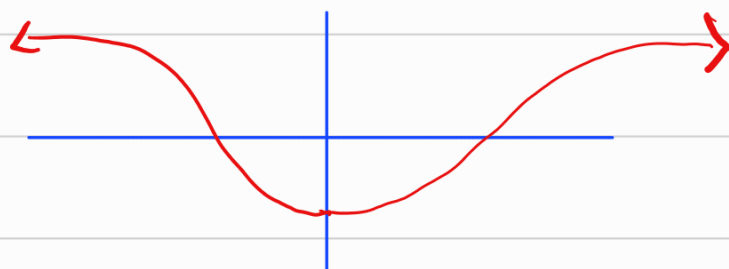
Earlier when we talked about limits we ran into Vertical asymptotes, these are values a where $\lim_{x \rightarrow a} f(x) = \pm \infty$. Potentially sided limits as well. What happened here was as x got arbitrarily close to a , the values of $f(x)$ became arbitrarily large.

In this section, we consider what happens to $f(x)$ as the input x becomes arbitrarily large. This called the end behavior.

Somewhat unintuitively, as $x \rightarrow \pm \infty$, the value of $f(x)$ can grow to infinity, converge to some finite value, or tend to no value at all. So anything is possible.

Ex: $f(x) = \frac{x^2 - 1}{x^2 + 1}$, Numerically

x	$f(x)$
0	-1
± 1	0
± 10	0.980198
± 100	0.9998
± 1000	0.999998



$$\lim_{x \rightarrow \pm \infty} f(x) = 1$$

Def:

- Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

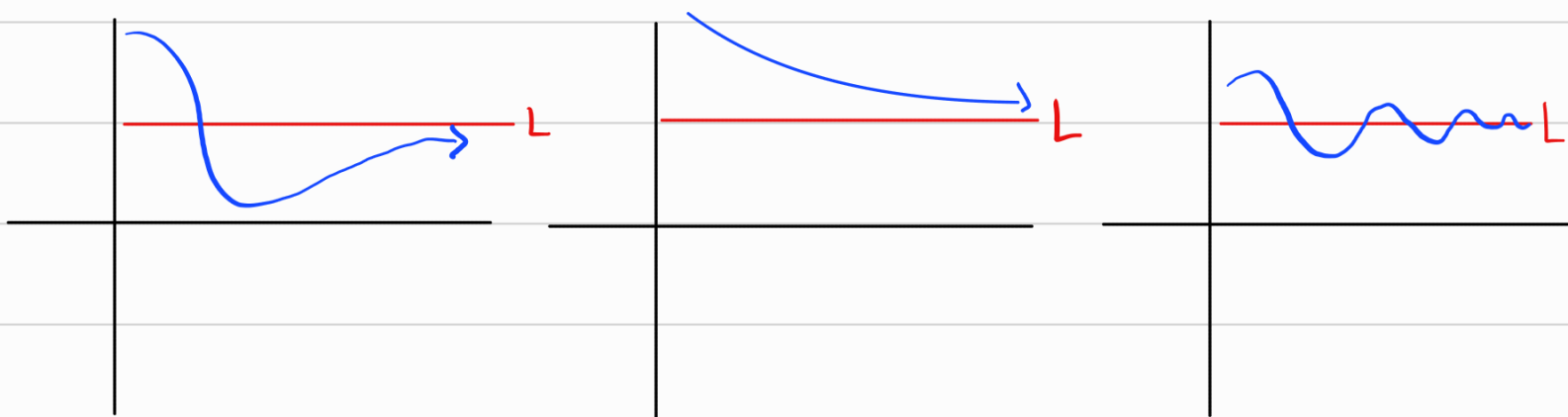
Means values of $f(x)$ can be made arbitrarily close to L by choosing x sufficiently large.

- Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

Means values of $f(x)$ can be made arbitrarily close to L by choosing x sufficiently large negative

There are many ways that limits at infinity can look



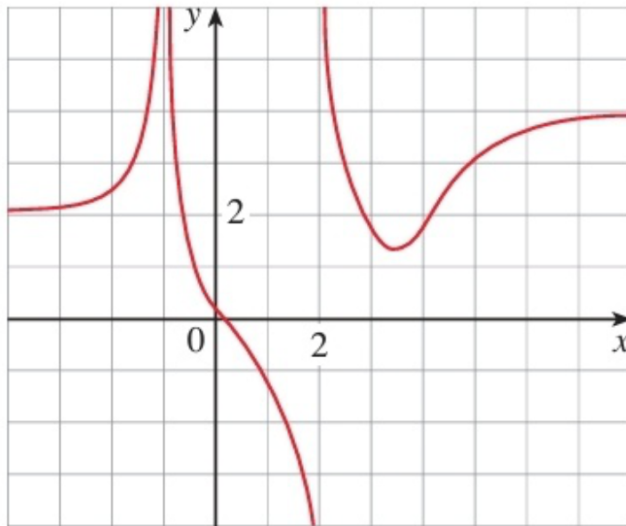
These all eventually go to L .

Def:

The line $y=L$ is called a horizontal Asymptote of the curve $y=f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Ex: Find all asymptotes of the following



$$\lim_{x \rightarrow -\infty} f(x) = 2, \quad \lim_{x \rightarrow \infty} f(x) = 3$$

$$\lim_{x \rightarrow -1} f(x) = \infty, \quad \lim_{x \rightarrow 2^-} f(x) = -\infty, \quad \lim_{x \rightarrow 2^+} f(x) = \infty$$

Vertical Asymptotes: $x = -1, x = 2$

Horizontal Asymptotes: $y = 2, y = 3$

Since we can't just evaluate $\lim_{x \rightarrow \infty} f(x)$ by direct substitution, we will have to do a lot of rewriting and utilize the following fact

Thm:

If $r > 0$ is a rational number, then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

Further, if x^r is defined for all x , then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0 \quad \text{as well}$$

Ex: Find Asymptotes of $f(x) = \frac{3x^2 - x - 2}{5x^2 + 4x + 1}$

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \quad \begin{array}{l} \text{highest power of } x \\ \text{in denominator} \end{array}$$

$$= \lim_{x \rightarrow \infty} \frac{3 - \frac{x}{x^2} - \frac{2}{x^2}}{5 + \frac{4x}{x^2} + \frac{1}{x^2}}$$

$$= \frac{\lim_{x \rightarrow \infty} \left(3 - \frac{1}{x} - \frac{2}{x^2} \right)}{\lim_{x \rightarrow \infty} \left(5 + \frac{4}{x} + \frac{1}{x^2} \right)}$$

$$= \frac{\lim_{x \rightarrow \infty} 3 - \lim_{x \rightarrow \infty} \frac{1}{x} - \lim_{x \rightarrow \infty} \frac{2}{x^2}}{\lim_{x \rightarrow \infty} 5 + \lim_{x \rightarrow \infty} \frac{4}{x} + \lim_{x \rightarrow \infty} \frac{1}{x^2}} = \frac{3}{5}$$

The $\lim_{x \rightarrow \infty} f(x)$ is the same in this case.

Horizontal Asymptote
 $y = \frac{3}{5}$

Ex: Find horizontal Asymptotes of $f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{2x^2+1}}{3x-5} \cdot \frac{\frac{1}{x}}{\frac{1}{x}} \quad \text{highest power of } x \text{ in denominator}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2+1}}{x}}{3 - \frac{5}{x}} \quad \sim \quad x = \sqrt{x^2} \quad \text{since } x > 0$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2+1}}{\sqrt{x^2}}}{3 - \frac{5}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{x^2}}}{3}$$

$$= \frac{\sqrt{2}}{3}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2+1}}{3x-5} \cdot \frac{\frac{1}{x}}{\frac{1}{x}}$$

$$\lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{2x^2+1}}{x}}{3 - \frac{5}{x}} \quad \sim \quad x = -\sqrt{x^2} \quad \text{since } x < 0$$

$$\lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{2x^2+1}}{-\sqrt{x^2}}}{3 - \frac{5}{x}}$$

$$\lim_{x \rightarrow -\infty} \frac{-\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}}$$

$$= -\frac{\sqrt{2}}{3}$$

Horizontal Asymptotes

$$y = \frac{\sqrt{2}}{3}, \quad y = -\frac{\sqrt{2}}{3}$$

When taking limits at infinity, we need to be careful only for forms of the type $\frac{\infty}{\infty}$ or $\infty - \infty$. In these cases, the limit can be any number.

Ex: $\lim_{x \rightarrow \infty} \sqrt{x^2+1} - x$

This is an $\infty - \infty$ case. We also see that as $x \rightarrow \infty$ $\sqrt{x^2+1}$ and x are similar. So we can't just evaluate straight away. We use an old trick

$$\lim_{x \rightarrow \infty} \sqrt{x^2+1} - x \cdot \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{x^2+1 - x^2}{\sqrt{x^2+1} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x^2+1} + x} \quad \rightarrow \text{as } x \rightarrow \infty, \sqrt{x^2+1} + x \rightarrow \infty \text{ as well.}$$

$$= 0$$

To simplify some limits, we make variable substitutions.

Ex: $\lim_{x \rightarrow \infty} \sin\left(\frac{1}{x}\right)$, let $t = \frac{1}{x}$ as $x \rightarrow \infty$, $\frac{1}{x} \rightarrow 0^+$

so $\downarrow$

$$\lim_{t \rightarrow 0^+} \sin(t) = \sin(0) = 0$$

Ex: $\lim_{x \rightarrow \infty} \sin(x)$ DNE

as $x \rightarrow \infty$, $\sin(x)$ will oscillate between -1 and 1 an infinite number of times, so it never goes to one single value. Thus, $\lim_{x \rightarrow \infty} \sin(x)$ DNE.

Infinite Limits at Infinity

To further understand the end-behavior of functions, we need to find away to address when $\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty$. That is, how do we know if as x becomes large, $f(x)$ does as well.

Ex: $\lim_{x \rightarrow \infty} x^3 = \infty$, $\lim_{x \rightarrow -\infty} x^3 = -\infty$

These are believable and we can see this by just evaluating at a points.

What becomes difficult is when we have addition and subtraction to consider. We often will need to rewrite the function.

Ex: $\lim_{x \rightarrow \infty} (x^2 - x)$

Since ∞ isn't a Number, we cannot use the usual limit laws.

Instead,

$$\lim_{x \rightarrow \infty} (x^2 - x) = \lim_{x \rightarrow \infty} x(x-1)$$

both x and $x-1$ become large as $x \rightarrow \infty$

So

$$\lim_{x \rightarrow \infty} x(x-1) = \infty$$

Ex:

$$\lim_{x \rightarrow \infty} \frac{x^2 + x}{3 - x} \cdot \frac{\frac{1}{x}}{\frac{1}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{x+1}{\frac{3}{x} - 1}$$

as $x \rightarrow \infty$, the denominator goes to -1
and, the numerator goes to ∞

So

$$\lim_{x \rightarrow \infty} \frac{x^2 + x}{3 - x} = -\infty$$

Ex: Quickly Sketch $y = (x-2)^4(x+1)^3(x-1)$.

We'll just look at intercepts and end behavior.

- $\lim_{x \rightarrow \infty} (x-2)^4(x+1)^3(x-1) = \infty$, as all 3 terms are positive
- $\lim_{x \rightarrow -\infty} (x-2)^4(x+1)^3(x-1) = \infty$, as 1 term is positive, the other 2 are negative.
- y intercept is
 $(0-2)^4(0+1)^3(0-1) = -16$

• x intercepts

$$x = -1, 1, 2$$



	$(x-2)^4$	$(x+1)^3$	$(x-1)$
$(-\infty, -1)$	+	-	-
$(-1, 1)$	+	+	-
$(1, 2)$	+	+	+
$(2, \infty)$	+	+	+

Function is negative on
 $(-1, 1)$

Positive everywhere else.